Junbo Li

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Portfolio & LinkedIn: https://github.com/Junbo345| https://www.linkedin.com/in/lijunboprofile/
Programming Languages: **Python, SQL**, **R**

EDUCATION

University of Waterloo Waterloo, ON | 09/2025 - 01/2027 (Expected)

Master of Data Science and Artificial Intelligence (CO-OP) (CGPA 4.00/4.00)

- Related coursework: **Advanced Machine Learning Systems, Deep Learning**, Probabilistic & Bayesian Methods, **Data Engineering (Spark), Database Systems (SQL), Exploratory Data Analysis**, Optimization for Data Science, introduction to AI.
- Expected coursework: **Research Topics in NLP**
- Awards: **Scotiabank Scholarship.**

University of Toronto Toronto, ON | 09/2021 - 04/2025

*Bachelor of Science, Double Major in Math and Statistics, with **High Distinction** (CGPA 3.98/4.00)*

- Related coursework: **Experimental Design (A/B Testing), Data Mining, Machine Learning, Statistical Modeling,** Business & Decision Analytics**,** Data Visualization, Time Series Analysis**,** Regression, Probability & Statistics
- Awards: Dean's List Scholar (2023, 2024, 2025); University of Toronto Excellence Award Recipient.

SKILLS

- **Machine Learning & Modeling:** PyTorch, Scikit-learn, GBM/XGBoost, feature engineering, model validation & performance evaluation, class-imbalanced learning, statistical modeling, time series analysis, EDA
- **Data Engineering & Pipelines:** Pandas, NumPy, Spark/PySpark, reproducible pipelines, data quality checks, experiment tracking
- **Software Engineering & ML Systems:** Git/GitHub (PR reviews & merges), Linux, REST APIs & model serving (FastAPI/Flask), structured experiment workflows, unit & validation testing, model debugging
- **Cloud, Visualization & Communication:** GCP, BigQuery, Matplotlib, Seaborn, Tableau, Power BI, Excel

WORK EXPERIENCE

University of Toronto Dunlap Institute of Astronomy Toronto, ON | 05/2024 - 05/2025 (1 yr)

Machine Learning Research Assistant | Machine learning, Python, SQL, Git

- Developed **deep learning models** to classify **100k+ galaxy images** from the Euclid telescope, reducing labeled data requirements by **60%** using **active learning** and aligning evaluation metrics with scientific objectives.
- Designed loss functions and metrics for **20:1 class imbalance**, improving **precision** from **30% to 70%** while maintaining recall.
- Built and maintained end-to-end training pipelines using **PyTorch Lightning** across multiple experimental iterations.
- Owned internal deployment workflows, including data preprocessing, inference validation, and post-training performance checks in **production-like research environments.**
- Open-sourced code on GitHub, actively reviewed PRs and approved merges, and delivered technical reports and presentations to **non-technical stakeholders**; work featured in university news.

Mater Kong Beijing, China| 06/2023 - 09/2023 (4 mos)

Field Data Collection Analyst | Data Analysis, Excel

- Monitored and maintained operational status of **50+ assets** ensuring **data accuracy** and compliance.
- Built and maintained **reporting workflows** using **Excel and SQL**, reducing issue resolution time by **40%**.
- Analyzed **usage patterns** and delivered **optimization recommendations**, reducing downtime by **20%**.
- Streamlined cross-team communication, reducing coordination time from **over 1 hour to less than 20 minutes**.

PROJECTS

Cyclic VAE–GAN for Image Generation — Research Project | Python, PyTorch 09/2025 – 12/2025(4 mos)

- Designed a **cyclic VAE–GAN training framework** that separates reconstruction and adversarial objectives, improving training stability and sample realism compared to standalone generative models.
- Achieved **55% improvement** in generative quality on MNIST, reducing FID from 23.5 (VAE) and 134.9 (GAN) to 10.4, and **35% FID reduction on Fashion-MNIST**.
- **Improved** discriminability of generated samples **by 40% ROC-AUC** in real-vs-generated classification tasks, indicating higher visual realism and diversity under fixed training budgets.

Exploring Determinants of Child Mortality — Global Data Analysis Project| Research Project | R, SQL, Git 09/2024 - 12/2024 (4mos)

- Defined research questions and integrated **multi-source datasets (100+ countries)** on mortality, vaccination, health spending, and food production using **SQL** and **World Bank APIs**.
- Built modular, reproducible **Bayesian regression pipeline in R**, validating results via posterior predictive checks.
- Identified significant drivers of child mortality and **translated results into actionable policy** insights for public health investment.